# Abhishek Yadav

**Backend & Distributed Systems Engineer | Java 17 & Microservices**

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## PROFESSIONAL SUMMARY

Software Development Engineer specializing in distributed systems, high-concurrency architectures, and low-latency data storage using Java, Python, FastAPI, and Spring Boot. Passionate about building robust APIs, optimizing system throughput, and automating production workflows with clean code and rigorous unit testing.

## WORK EXPERIENCE

**Software Development Engineer (SDE) at Anantasutra** May 2026 – Nov 2026

- Architected and deployed high-throughput Python data pipelines and RESTful microservices using **BeautifulSoup, Requests, and FastAPI**, boosting data ingestion by **40%** with automated verification and deduplication.
- Engineered automated **GitHub Actions CI/CD** pipelines and comprehensive unit test suites, reducing invalid transaction records by **30%** and ensuring zero-downtime microservice deployments.
- Implemented resilient error handling and asynchronous queueing with Celery and Redis, improving backend endpoint throughput under peak traffic loads.

**Data Engineer Intern at Anantasutra** Nov 2025 – Jan 2026

- Built automated lead-generation and web scraping pipelines with Python and RESTful API integrations, increasing data acquisition by 40% while improving data quality by 35% through backend verification.
- Developed microservices-based automation with GitHub Actions CI/CD and unit testing, improving overall data pipeline reliability and fault tolerance.

## TECHNICAL PROJECTS

**Distributed Key-Value Store (Java 17)** github.com/Abhiboss07/Distributed-KV-Store

- Architected a high-concurrency in-memory KV store in Java 17 featuring a multi-client TCP server (FixedThreadPool) and O(1) LRU cache eviction.
- Implemented dual-strategy TTL expiry and AOF write-ahead logging with fsync policies and atomic compaction; verified via 144 unit and concurrency tests.

**Voice AI SaaS Platform — Full-Stack Agentic Telephony** github.com/Abhiboss07/Voice-Agent

- Built a cloud-native Voice AI SaaS platform with microservices backend (FastAPI, Redis, Celery, Docker), LiveKit SIP telephony, and OpenAI/Gemini voice models.
- Engineered WebSocket real-time metrics streaming, JWT authorization, and responsive React/TypeScript UI for low-latency interactive audio sessions.

**Nexus Control Center — Linux System Telemetry & Diagnostics** github.com/Abhiboss07/Nexus

- Developed an open-source cross-platform system telemetry and diagnostics control center using Rust, Tauri v2, React, and TypeScript.
- Designed modular plugin architecture with real-time process monitoring, memory analysis, and AI-assisted performance diagnostics.

## TECHNICAL SKILLS

**Languages:** Java 17, Python, TypeScript, SQL, Rust, C++

**Backend & Frameworks:** FastAPI, Spring Boot, RESTful APIs, Microservices, gRPC

**Cloud, CI/CD & DevOps:** Docker, CI/CD, GitHub Actions, Linux, Kubernetes Basics

**Distributed Systems & Storage:** Redis, MongoDB, MySQL, PostgreSQL, Celery

**APIs, Security & Protocols:** RESTful API Design, WebSockets, JWT, RBAC, OAuth2

**Testing & Reliability:** Unit & Concurrency Testing (JUnit 5, PyTest), TDD

**AI & Agentic Systems:** OpenAI, Gemini, Agentic Workflows, Prompt Engineering

**Frontend & UI:** React, TypeScript, Tailwind CSS, HTML5/CSS3

**Data Engineering & ETL:** Data Pipelines, Web Scraping, Process Automation, ETL

**Developer Tools:** Git, VS Code, Cursor, Postman, Linux (Arch/CachyOS)

## EDUCATION & COURSEWORK

**Bachelor of Technology in Computer Science & Engineering** 2023 – 2027 (Expected)

Veer Bahadur Singh Purvanchal University

**Relevant Coursework:** Distributed Systems, Object-Oriented Programming (Java), Data Structures & Algorithms, Computer Networks, Operating Systems, Database Management Systems

## CERTIFICATIONS & HONORS

- **Deloitte Australia Technology Simulation:** Software Architecture, Cloud Infrastructure, Telemetry & Code Review (Forage).
- **CodSoft & Cognifyz Technologies:** Verified Java Software Development Certification, Advanced Object-Oriented Design.
- **Innovation Award:** Selected Participant & Pitch Presenter at Bharat Innovation Conclave.